RTDL Embree Paper-Style Report

Abstract

? RTDL is a Python-hosted DSL and compiler/runtime stack for non-graphics ray tracing workloads, with RayJoin as the target problem family. This report consolidates the current Embree-phase evidence into a single paper-style artifact.

? The present system supports six audited workload families, executes them through a Python CPU reference and a native Embree backend, and now includes a scaled Section 5.6 scalability analogue for lsi and pip.

? The resulting report is intentionally scoped as an Embree baseline, not as the final OptiX/NVIDIA v0.1 reproduction.

Current contribution

? Executable DSL surface: lsi, pip, overlay, ray_tri_hitcount, segment_polygon_hitcount, and point_nearest_segment.

? Audited local backends: run_cpu(...) for semantic reference and run_embree(...) for native execution.

? Paper-facing status: Figure 13 and Figure 14 analogues are implemented; Table 3, Table 4, and Figure 15 analogue remain open.

Generated from checked-in RTDL benchmark and report artifacts on 2026-03-31.

Method and Scope

Runtime and validation contract

? Every benchmarked case in the current baseline is parity-checked between run_cpu(...) and run_embree(...) before timing claims are accepted.

? The current precision model is float_approx only; the report does not claim robust or exact computational geometry.

? The overlay workload remains an overlay-seed analogue rather than full polygon materialization.

Paper-reproduction boundary

? Figure 13 and Figure 14 are supported through deterministic synthetic generators and a fixed-R/varying-S Embree experiment runner.

? The current analogue uses R=800 and S=160..800 polygons under uniform and gaussian distributions, rather than the paper's RT-core scale.

? Table 3 and Table 4 still require larger paper-aligned dataset acquisition and conversion, and Figure 15 still lacks a matching RTDL overlay-speedup generator.

Embree Baseline Results

Evaluation matrix

? Total current benchmark cases: 17 with all parity checks passing = True.

? Best observed Embree speedup vs CPU: ray_synthetic_large at 165.14x.

? Slowest Embree case by mean latency: ray_synthetic_large at 0.001822s.

Largest currently evaluated cases

? lsi: lsi_county_tiled_x8 at 0.000413s mean.

? pip: pip_county_tiled_x8 at 0.000450s mean.

? overlay: overlay_county_soil_tiled_x8 at 0.000614s mean.

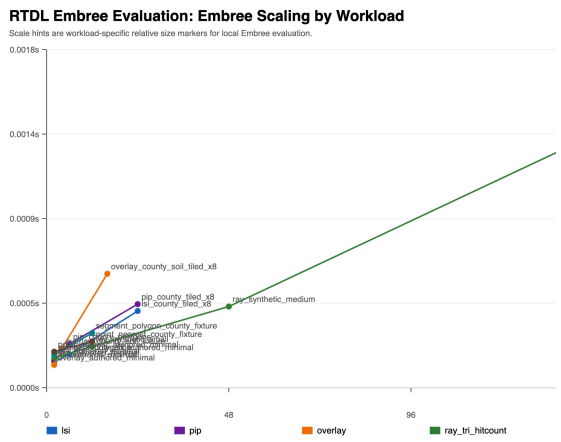
? ray_tri_hitcount: ray_synthetic_large at 0.001822s mean.

? segment_polygon_hitcount: segment_polygon_county_fixture at 0.000294s mean.

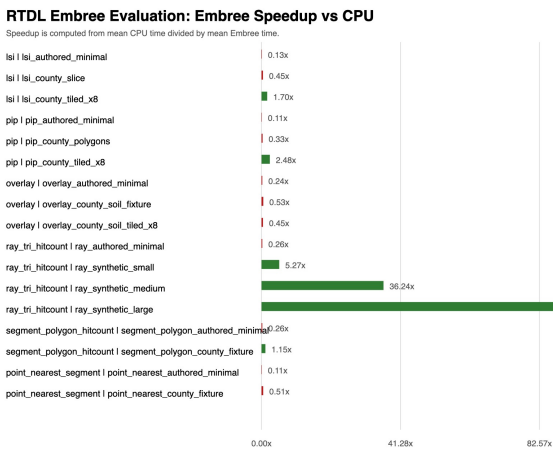
? point_nearest_segment: point_nearest_county_fixture at 0.000249s mean.

Baseline Figures

RTDL scaling-by-workload figure



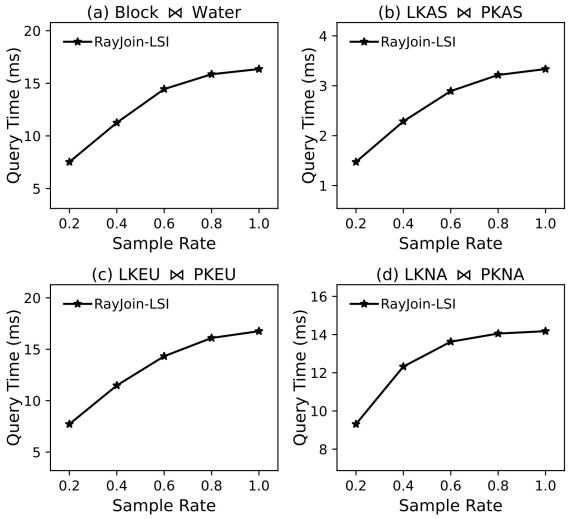
RTDL speedup-by-case figure



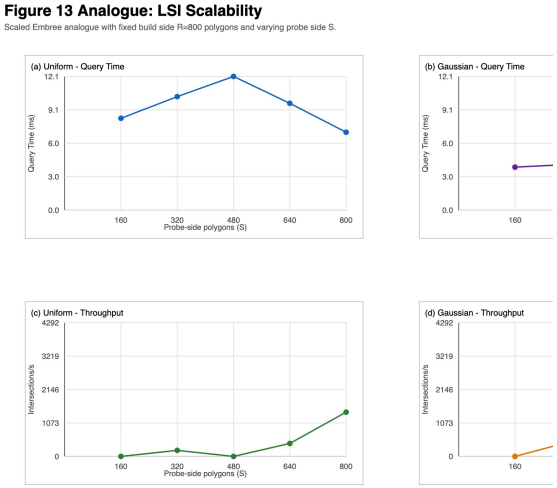
These figures come from the frozen Embree baseline evaluation matrix. They summarize the current six-workload local benchmark surface before the paper-specific reproduction steps. They are not direct RayJoin paper figures, but they provide the current baseline from which the paper-reproduction phase is extending.

Figure 13 Correspondence

RayJoin paper Figure 13 reference



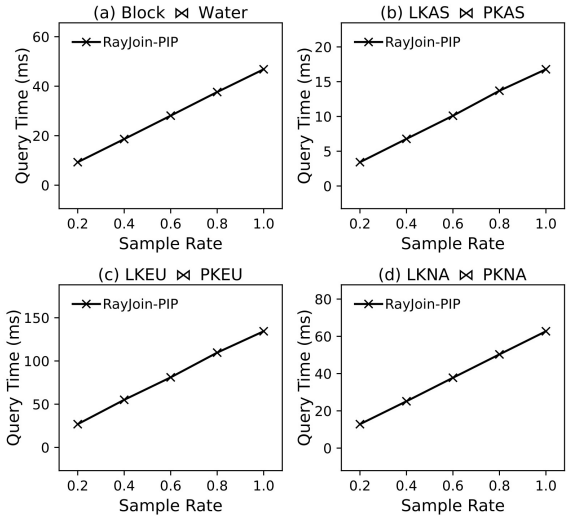
RTDL Embree Figure 13 analogue



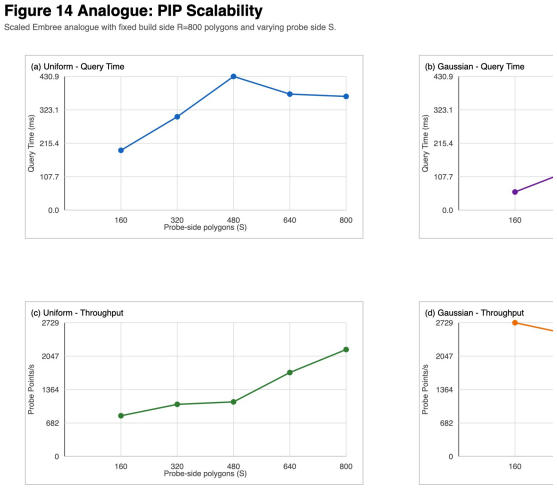
The left image is the corresponding RayJoin paper scalability figure for LSI. The right image is RTDL's current Embree-phase analogue.
RTDL now supports deterministic uniform and gaussian synthetic LSI scaling runs with query-time and throughput curves, but at a smaller fixed-R/varying-S scale than the paper.

Figure 14 Correspondence

RayJoin paper Figure 14 reference



RTDL Embree Figure 14 analogue



The left image is the corresponding RayJoin paper scalability figure for PIP. The right image is RTDL's current Embree-phase analogue.
For PIP, throughput is reported as probe-points per second. The analogue is correctness-gated with CPU-vs-Embree parity on reduced representative samples.

Section 5.6 Quantitative Highlights

Correctness gate

? All reduced parity checks for the Section 5.6 analogue passed: True.

Representative throughput trends

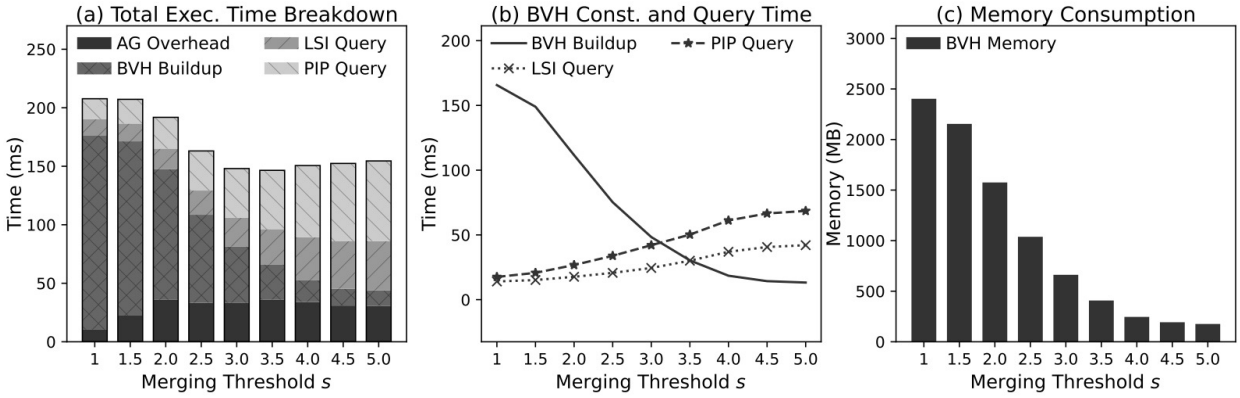
? Uniform LSI throughput: 160 probe polygons -> 0.00 intersections/s, rising to 800 -> 1421.33 intersections/s.

? Gaussian LSI throughput: 320 -> 483.17 intersections/s, rising to 800 -> 4292.25 intersections/s.

? Uniform PIP throughput: 160 -> 829.91 probe-points/s, rising to 800 -> 2182.83 probe-points/s.

? Gaussian PIP throughput: 160 -> 2729.58 probe-points/s, ending at 800 -> 1970.94 probe-points/s.

Figure 15 Current Status



This page includes the corresponding RayJoin paper overlay-speedup figure as a reference target.

RTDL does not yet have the matching Figure 15 analogue. Table 4 and Figure 15 remain open because the current overlay path is an overlay-seed analogue and the larger paper-aligned dataset pairs are not all integrated yet.

Discussion and Next Steps

What this report establishes

? RTDL is now beyond a syntax-only prototype: it has a validated local backend, reproducible benchmark/report generation, and initial paper-structured experiment support on Embree.

? The Section 5.6 slice is implemented and accepted as an Embree-phase analogue, which reduces risk for the later OptiX/NVIDIA phase.

What remains for full paper-style reproduction

? Acquire and normalize more of the RayJoin Table 3 and Table 4 dataset pairs into the RTDL pipeline.

? Add larger paper-structured cases to the Embree evaluation matrix with CPU-vs-Embree parity gates.

? Implement the RTDL overlay-speedup analogue needed for Figure 15.

? Carry the same workload and figure pipeline forward to the eventual NVIDIA/OptiX backend.